

Experiment 'p_6_2_a_swap_first_last_array' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 2

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 2

Problem Specification

Problem name: p_6_2_a_swap_first_last_array

Natural language statement: Write a method to swap the first and last elements in an array.

Method signature: p_6_2_a_swap_first_last_array(inputs: array<int>) returns (result: array<int>)

Ensures

- `result.Length == inputs.Length`
- `result[0] == inputs[inputs.Length - 1]`
- `result[inputs.Length - 1] == inputs[0]`
- forall `i :: 1 <= i < inputs.Length - 1 ==> result[i] == inputs[i]`

Requires

- `inputs.Length >= 2`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to swap the first and last elements in an array.

The signature should be:

`p_6_2_a_swap_first_last_array(inputs: array<int>) returns (result: array<int>)`

The method should respect the following contract:

`requires inputs.Length >= 2, ensures result.Length == inputs.Length, ensures result[0] ==`
 ↪ `inputs[inputs.Length - 1], ensures result[inputs.Length - 1] == inputs[0],`
 ↪ `ensures forall i :: 1 <= i < inputs.Length - 1 ==> result[i] == inputs[i]`

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn0BYb8uF49vGABxwtCnM28sfJMzV

Model: gpt-5.5-2026-04-23

Created at: 1780571156

Finish reason: stop

Usage: 378 tokens in, and 587 tokens out

```
{
  result := new int[inputs.Length](i =>
    if i == 0 then inputs[inputs.Length - 1]
    else if i == inputs.Length - 1 then inputs[0]
    else inputs[i]
  );
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```
problem_data/p_6_2_a_swap_first_last_array/program.dfy(9,25): Error: insufficient reads
    ↪ clause to read array element; Consider extracting inputs[inputs.Length - 1] to a
    ↪ local variable before the lambda expression, or adding 'reads inputs' in the
    ↪ enclosing lambda specification for resolution
|
9 | if i == 0 then inputs[inputs.Length - 1]
```

```

| ^
problem_data/p_6_2_a_swap_first_last_array/program.dfy(10,46): Error: insufficient reads
  ↳ clause to read array element; Consider extracting inputs[0] to a local variable
  ↳ before the lambda expression, or adding 'reads inputs' in the enclosing lambda
  ↳ specification for resolution
|
10 | else if i == inputs.Length - 1 then inputs[0]
| ^

problem_data/p_6_2_a_swap_first_last_array/program.dfy(11,15): Error: index out of range
|
11 | else inputs[i]
| ^

problem_data/p_6_2_a_swap_first_last_array/program.dfy(11,15): Error: insufficient reads
  ↳ clause to read array element; Consider extracting inputs[i] to a local variable
  ↳ before the lambda expression, or adding 'reads inputs' in the enclosing lambda
  ↳ specification for resolution
|
11 | else inputs[i]
| ^

Dafny program verifier finished with 1 verified, 4 errors

Correct the error in code by altering only the method body. Produce and show only the
  ↳ Dafny body, including the curly braces that surround it. Do not show the signature
  ↳ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

```

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn0C9ABvC7L1AFVIQcCciiqw1vpyC

Model: gpt-5.5-2026-04-23

Created at: 1780571193

Finish reason: stop

Usage: 1017 tokens in, and 893 tokens out

```

{
  result := new int[inputs.Length];
  result[0] := inputs[inputs.Length - 1];
  result[inputs.Length - 1] := inputs[0];

  var i := 1;
  while i < inputs.Length - 1
    invariant 1 <= i <= inputs.Length - 1
    invariant result.Length == inputs.Length
    invariant result[0] == inputs[inputs.Length - 1]
    invariant result[inputs.Length - 1] == inputs[0]
    invariant forall j :: 1 <= j < i ==> result[j] == inputs[j]

```

```

{
  result[i] := inputs[i];
  i := i + 1;
}
}

```

Final Program

```

method p_6_2_a_swap_first_last_array(inputs: array<int>) returns (result: array<int>)
  requires inputs.Length >= 2
  ensures result.Length == inputs.Length
  ensures result[0] == inputs[inputs.Length - 1]
  ensures result[inputs.Length - 1] == inputs[0]
  ensures forall i :: 1 <= i < inputs.Length - 1 ==> result[i] == inputs[i]
{
  result := new int[inputs.Length];
  result[0] := inputs[inputs.Length - 1];
  result[inputs.Length - 1] := inputs[0];

  var i := 1;
  while i < inputs.Length - 1
    invariant 1 <= i <= inputs.Length - 1
    invariant result.Length == inputs.Length
    invariant result[0] == inputs[inputs.Length - 1]
    invariant result[inputs.Length - 1] == inputs[0]
    invariant forall j :: 1 <= j < i ==> result[j] == inputs[j]
    {
      result[i] := inputs[i];
      i := i + 1;
    }
}

```

Total Token Usage

Input tokens: 1395

Output tokens: 1480

Reasoning tokens: 1219

Sum of 'total tokens': 2875

Experiment Timings

Overall Experiment started at 1780571156024, ended at 1780571228535, lasting 72511ms (72.51 seconds)

Iteration #1 started at 1780571156025, ended at 1780571192732, lasting 36707ms (36.71 seconds)

Iteration #2 started at 1780571192733, ended at 1780571228534, lasting 35801ms (35.80 seconds)